

Editorial

After well over a decade of liberalisation across a growing number of countries world-wide, the telecommunications market is becoming ever more complex as new operators jostle for position among incumbent operators, and even newer companies exploit other niches in the market-place. Access technology is always a dominant part of the planning process since it generally forms one of the most significant investments required, as well as offering competitive advantage for certain services. The choice of technology is therefore of critical concern to new and old operating companies.

In this timely special edition, the papers are grouped into a number of theme areas. The first two papers by France et al and Adams et al set the scene by providing an introduction to the market influences on access networks and the technological advances which will help to meet increasingly demanding customer requirements. These two papers also act as a summary of the most significant access technologies currently in use by the most significant access network providers (i.e. ex-incumbent operators).

The next six papers focus on wireline access networks. Foster et al discuss digital subscriber line (DSL) technologies which dramatically enhance the bandwidths which can be delivered to customers over the existing copper infrastructure; an application of these technologies is given in the paper on Midband by Dickie and Mackenzie. A strategy for broadband access, the Full Service Access Network (FSAN), is described in Quayle et al, and the extension of these concepts into the home environment ('the last few metres') is given in Hughes and Thorne. An overview of cable TV networks and their capabilities are given in Jewell et al. The final paper in this grouping addresses the prospects of reductions in cost for optical fibre installation in the access environment.

Wireless technologies form the next grouping, with papers on point-to-multipoint fixed radio access by Merrett et al and the extensive coverage provided by satellite technologies by Wakeling et al.

Having completed a review of access transmission technologies, the next two papers review opportunities for new entrants. Searle et al provide an overview of access strategies open to new operators, while Mallett and Fletcher focus on indirect access obtained via interconnection with other network operators' facilities.

The following five papers move on to operational aspects of the access network. Asumu et al present a discussion of automated planning for wireline access networks. Papers by Trigger/Cameron and Bell/Trigger provide two perspectives on current account cost reduction in the access network. The final two papers in this grouping, by Ford and by Redstall et al, provide a detailed analysis of equipment and pole reliability.

Discerning network planners, while trying to make the optimum choice from the bewildering array of access network technologies before them, must also keep an eye on the future. They must satisfy themselves that their choice will not become obsolete before the appropriate time. To that end the final paper in this special edition provides a thought-provoking insight into future access networks.

Finally, our thanks to all the contributors to this edition of the Journal who have produced a most comprehensive summary of the dynamic developments at the access end of the network.

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